

Aditya Bidappa M V

cybersecurity researcher · full-stack security builder · zero-trust systems · ML for defensive security

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B.Tech Cybersecurity · Dayananda Sagar University

CONTACT

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SKILLS

LANGUAGES

Python C SQL Bash TypeScript
HTML LaTeX

FRAMEWORKS

FastAPI Express React
React Native Expo

DATA

PostgreSQL MySQL JSONB MongoDB

SECURITY & STANDARDS

Zero Trust NIST SP 800-207
CISA ZTMM v2
OWASP Testing Guide v4
IDOR & Auth-Bypass Testing
Blind & Error-Based SQLi

PENTESTING TOOLKIT

Burp Suite CE OWASP ZAP
Hoppscotch gobuster nuclei
subfinder nmap Wireshark
SpiderFoot Autopsy Kali Linux

MACHINE LEARNING

XGBoost scikit-learn
Random Forest Isolation Forest
Multi-Transformer Ensembles

AUTOMATION & REPORTING

openpyxl Google Apps Script
GitHub Actions Typst

TOOLING

Git GitHub

AI – LLM ASSISTANTS

Claude ChatGPT Gemini DeepSeek
Perplexity

AI – VOICE / AUDIO

Whisper

AI – AGENTIC / AUTOMATION

n8n OpenRouter

PROFESSIONAL SUMMARY

I'm a second-year Cybersecurity undergrad who treats research and engineering as the same discipline. My work moves between reading standards (NIST SP 800-207, CISA ZTMM v2) and turning them into things that run – CLI tools, deployed web apps, mobile auditors.

The through-line is measurable trust: scoring how exposed a network is, grading an organisation's zero-trust maturity, quantifying how likely a prompt is to be an injection. I care about correctness over shine, and about being honest when a metric is too good to be true.

PROJECTS

PromptGuard

FastAPI · React · TypeScript · XGBoost · scikit-learn

Prompt-injection risk scorer for LLM apps

- ▶ FastAPI risk-scoring service with a React + TypeScript frontend and an XGBoost / scikit-learn backend.
- ▶ Trained on Hugging Face prompt-injection data plus synthetic adversarial prompts generated locally with DeepSeek-R1 1.5B on Ollama.

87.08% accuracy 85% injection recall 95% precision

<https://github.com/BrownJesus69/promptguard>

SME-ZT CLI

Python CLI · CISA ZTMM v2 · NIST SP 800-207 · CSA SMB

Zero-trust maturity assessment for small & medium enterprises

- ▶ A five-pillar, five-level maturity model aligned to CISA ZTMM v2, NIST SP 800-207 and CSA SMB guidance.
- ▶ Weighted scoring with Level-1 capping logic to stop over-scoring, plus detailed textual self-assessment reports.
- ▶ Validated against three documented SME breach case studies and folded into the published manuscript.

<https://github.com/BrownJesus69/zt-maturity-sme>

MUN Argument Builder LIVE

FastAPI · Groq API · Llama 3.1 70B · Render

Deployed argument engine for Model UN delegates

- ▶ Generates structured arguments, counter-arguments, evidence prompts and fallacy checks on demand.
- ▶ FastAPI backend on Groq's Llama 3.1 70B, with GitHub-triggered auto-deploy on Render.
- ▶ Custom domain via GoDaddy and TLS via Let's Encrypt.

<https://quantumpandits.com> <https://github.com/BrownJesus69/mun-argument-builder>

NetworkAudit

React Native · Expo SDK 54 · TypeScript · Express · Zod · Neon PostgreSQL

Real-time Wi-Fi & hotspot security auditor for Android

- ▶ Scans the connected network live and computes a 0–100 security score with an A–F grade.
- ▶ Flags CVE-based findings: default gateway IPs, single DNS server, external-DNS logging risk and SSID identity leakage.
- ▶ Full-stack: RN + Expo client, Express + Zod API, Neon PostgreSQL, EAS Build on Render.

<https://github.com/BrownJesus69/networkaudit>

NutriLog

PostgreSQL · BCNF schema · Frontend · Docs

Relational dietary-logging system

- ▶ Owned the frontend, documentation and application logic for a PostgreSQL nutrition tracker.
- ▶ Correctness-oriented design grounded in a BCNF-normalised schema.

<https://github.com/sampratigaurav/nutrilog>

EXPERIENCE

AI – SECURITY RESEARCH

- HexStrike AI
- MCP Server Orchestration

AI – AGENT FRAMEWORKS

- Z.AI
- Hermes AI
- OpenClaw

EDUCATION

B.Tech in Cybersecurity

Dayananda Sagar University, Bengaluru
Aug 2024 – Jun 2028 (Expected)
CGPA 7.5 / 10.0

Coursework: Introduction to Cybersecurity, AI in Cybersecurity, DBMS, Computer Networks

CERTIFICATIONS

Introduction to Cybersecurity

Cisco Networking Academy · Feb 2026

Ethical Hacker

Cisco Networking Academy · Apr 2026

ROLES & ACTIVITIES

President

DSU MUN Society

Joint Treasurer

Cybersecurity Club

Special Advisor

Debate Society

Secretary General

TNT MUN

Director General

COPE II

Rapporteur

DSUMUN II

Cybersecurity Intern

TabNet Security – Arova Fitness

Jun 2026 – Aug 2026

- ▶ Conducted authorized pentest on arovafitness.com and REST API (api.arovafitness.com) using Kali Linux and Burp Suite Community Edition – DNS/WHOIS recon, TLS review, HTTP header analysis, gobuster directory enumeration, and JavaScript bundle reverse-engineering to uncover an Express.js backend behind Cloudflare and map CORS preflight exposure on authentication-gated endpoints.
- ▶ Parsed OpenAPI specification to catalog 86+ production endpoints across External API, Gym Panel API, and Fitness API services (73 JWT-authenticated, 18 unauthenticated, 22 AI-backed); produced a 352-route API surface map and structured findings reports covering open ports, service versions, and endpoint-level vulnerabilities.
- ▶ Completed two assessment phases yielding 16 total findings (5 Phase 1 unauthenticated, 11 Phase 2 authenticated), including two Critical-severity vulnerabilities – premium plan escalation via PUT / api/user/plan and OTP invalidation flaws – validated via IDOR-style token-swapping and Hoppscotch functional test cases.
- ▶ Architected 192-row exercise library (100 Warmup + 92 Cardio, 12-column schema); built Python/openpyxl data pipeline and Google Apps Script automation to bulk-rename and organize image assets in Drive with lowercase-underscore-numbered convention, validating the platform's live admin category system.

RESEARCH & PUBLICATIONS

Zero Trust Maturity Assessment Framework for Small and Medium Enterprises (SME-ZT)

Co-author

A five-pillar, five-level maturity framework with SME-specific weighting and Level-1 capping to prevent over-scoring, validated across three breach case studies.

Security Threats and Countermeasures in IoMT ML-driven Healthcare Systems

Co-author

Survey of ML-based IoMT intrusion detection, federated learning, privacy preservation and layered defences against malware, DoS and adversarial attacks.

Deepfake Media Analysis: A Survey on Detection Techniques and Open Issues

Co-author

Review of deepfake generation methods, benchmark datasets, detection techniques and the surrounding ethical and legal frameworks.

A Baseline Simulation Framework for DDoS Attack Detection in IoMT Networks

Lead author

Wrote and formatted the full manuscript; showed that 'perfect' metrics on synthetic data arise from trivial one-dimensional separability in the SYN ratio.

AWARDS

- Best Delegate: Next Debate MUN'25 · UNSC, Australia
- High Commendation: Avent · UNSC, USA
- High Commendation: BITS Goa MUN'26 · UNSC, France
- Special Mention: MAHE MUN'25 · UNSC, United Kingdom
- Verbal Mention: DSBAMUN2 · UNSC, Russia
- Honorable Mention: DSU MUN Conference